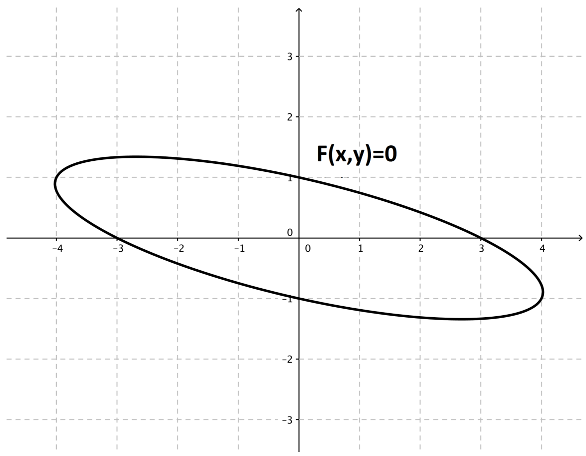
Problem 2

Problem 2

Problem 1 *Consider the equation $x^2 + 4xy + 9y^2 = 9$. Note: This equation is equivalent to $x^2 + 4xy + 9y^2 - 9 = 0$. Therefore it has a form $F(x, y) = 0$*



(a) *Find* $\dfrac{dy}{dx}$.

Explanation.

$$\frac{d}{dx}(x^2 + 4xy + 9y^2) = \frac{d}{dx}(9)$$

$$2x + \left(4y + 4x\frac{dy}{dx}\right) + 18y\frac{dy}{dx} = 0$$

$$4x\frac{dy}{dx} + 18y\frac{dy}{dx} = -2x - 4y$$

$$(4x + 18y)\frac{dy}{dx} = -2x - 4y$$

$$\frac{dy}{dx} = \frac{-2x - 4y}{4x + 18y}$$

provided that $4x + 18y \neq 0$.

(b) *Find the equation(s) of the tangent line(s) when $x = 0$. Draw the tangent line(s) on the above picture.*

Explanation. Plugging into the original equation, we see that when $x = 0$ we have that

$$0^2 + 4(0)y + 9y^2 = 9 \qquad \Longrightarrow \qquad y^2 = 1 \qquad \Longrightarrow \qquad y = \pm 1$$

Then,

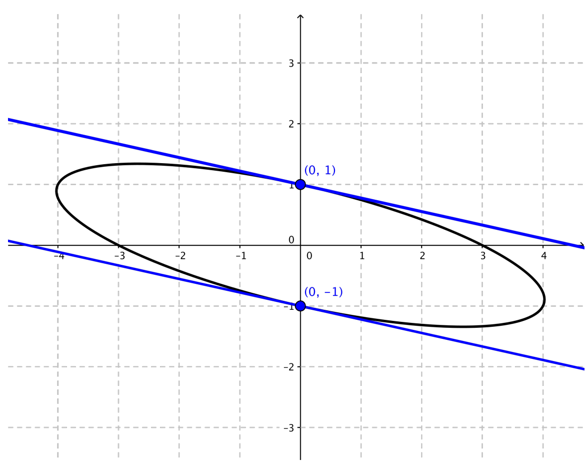
$$\left[\frac{dy}{dx}\right]_{(0,1)} = \frac{-4}{18} = -\frac{2}{9}$$

$$\left[\frac{dy}{dx}\right]_{(0,-1)} = \frac{4}{-18} = -\frac{2}{9}$$

So there are two tangent lines to the graph of the given equation when $x = 0$. The two points are $(0, 1)$ and $(0, -1)$, and the tangent lines at both points have slope $-\frac{2}{9}$. Thus, the equations of the two tangent lines are

$$y - 1 = -\frac{2}{9}(x - 0) \qquad \Longrightarrow \qquad y = -\frac{2}{9}x + 1$$

$$y + 1 = -\frac{2}{9}(x - 0) \qquad \Longrightarrow \qquad y = -\frac{2}{9}x - 1$$



(c) *Find the point(s) where the tangent line is horizontal. Draw the point(s) and line(s) on the above picture.*

Explanation. A line is horizontal if and only if its slope is 0. So we are looking for the points (x_0, y_0) such that $\left[\frac{dy}{dx}\right]_{(x_0,y_0)} = 0$. So we solve:

$$\frac{-2x - 4y}{4x + 18y} = 0 \qquad \Longrightarrow \qquad -2x - 4y = 0 \qquad \Longrightarrow \qquad x = -2y$$

But we also need the point to lie on the the given curve. So, letting $x = -2y$, we solve:

$$\begin{aligned} x^2 + 4xy + 9y^2 &= 9 \\ (-2y)^2 + 4(-2y)y + 9y^2 &= 9 \\ 4y^2 - 8y^2 + 9y^2 &= 9 \\ 5y^2 &= 9 \\ y^2 &= \frac{9}{5} \\ y &= \pm\frac{3}{\sqrt{5}} \end{aligned}$$

So there exists two solutions to the given equation where the tangent line to the graph has 0 slope. Those two points are $\left(-\frac{6}{\sqrt{5}}, \frac{3}{\sqrt{5}}\right)$ and $\left(\frac{6}{\sqrt{5}}, -\frac{3}{\sqrt{5}}\right)$.